

CRAF: CROSS-VIEW RESIDUAL-AWARE FUSION FOR DEEPPAKE SPEECH DETECTION

Minh-Xuan Phan[✉], Khalid Zaman[✉], Candy Olivia Mawalim[✉], Masashi Unoki[✉]

Japan Advanced Institute of Science and Technology

ABSTRACT

Recent advances in speech synthesis and voice conversion have made deepfake speech increasingly realistic, making generalization to unseen spoofing attacks a critical challenge. Pretrained speech and audio models offer a promising direction for improving robustness to such unseen attacks. Self-supervised learning (SSL) models capture fine-grained, low-level acoustic characteristics, whereas Auditory Large Language Models (ALLMs) provide higher-level contextual representations. These complementary views can provide useful cues for improving generalization to unseen attacks. However, direct fusion does not explicitly disentangle information shared across the two views from view-specific complementary information, limiting effective cross-view integration. To address this, we propose CRAF, a cross-view residual-aware fusion framework that uses ALLM-guided cross-view attention to enrich SSL representations and adopts ALLM as a high-level reference to separate ALLM-explainable information from complementary SSL residual information. The residual is selectively refined through adaptive gating and integrated through SSL-primary fusion. Experiments on ASVspoof 5 show that CRAF with Kimi-Audio achieves an EER of 5.96% and a minDCF of 0.1192, demonstrating robustness to unseen spoofing attacks.

Index Terms— Self-supervised learning, large language model, deepfake speech detection.

1. INTRODUCTION

Recent advances in speech synthesis and voice conversion have made deepfake speech increasingly realistic, posing growing risks to voice authentication and digital forensics [1, 2]. Deepfake detectors have evolved from handcrafted spectral and phase-based features [3] to end-to-end architectures such as RawNet2 [4] and AASIST [5]. Despite strong performance under matched conditions, generalization to unseen synthesis methods, recording conditions, and datasets remains challenging [6, 7, 8]. This motivates representations that capture transferable spoofing characteristics rather than attack- or dataset-specific patterns.

SSL models have become strong front-ends for deepfake speech detection because their learned representations provide informative features that are effective for distinguishing genuine and synthesized speech [9, 10]. However, although

SSL representations capture rich acoustic and contextual information, they are not explicitly designed to model the high-level semantic knowledge available from audio large language models. Recent studies have therefore explored ALLMs for deepfake detection [11, 12]. LLM representations provide rich semantic and contextual abstractions, but their semantic-dominant high-level representations overlook fine-grained acoustic cues important for spoof detection [13]. Textual acoustic grounding further highlights this limitation by explicitly incorporating acoustic descriptors into LLM-based detectors to bridge the gap between subtle acoustic information and the LLM semantic space [14]. Thus, SSL and ALLM provide complementary views: SSL preserves fine-grained acoustic characteristics, whereas ALLM provides higher-level contextual guidance.

The challenge is to exploit this complementarity without weakening acoustic information required for spoof detection. A straightforward approach is to concatenate or directly fuse SSL and ALLM representations. However, direct fusion treats both views uniformly and does not distinguish shared information from complementary SSL-specific characteristics [15]. This can introduce redundancy while reducing emphasis on fine-grained SSL information weakly represented by the ALLM [16]. Thus, the problem is not simply to combine SSL and ALLM representations, but to incorporate high-level guidance while preserving complementary SSL information.

To address this problem, we propose CRAF, a cross-view residual-aware fusion framework for deepfake speech detection. CRAF adopts an asymmetric design in which SSL serves as the primary representation and ALLM provides high-level guidance. First, ALLM-guided cross-view learning enriches the SSL representation with contextual information. CRAF then estimates SSL information recoverable under ALLM guidance and models the remaining SSL-specific information as a cross-view residual. The residual is adaptively refined to preserve complementary fine-grained characteristics. Finally, SSL-primary fusion selectively integrates guided and residual representations while retaining the original SSL stream.

The main contributions are threefold. First, we introduce an asymmetric SSL–ALLM formulation that exploits high-level ALLM guidance while retaining SSL as the primary fine-grained representation. Second, we propose ALLM-

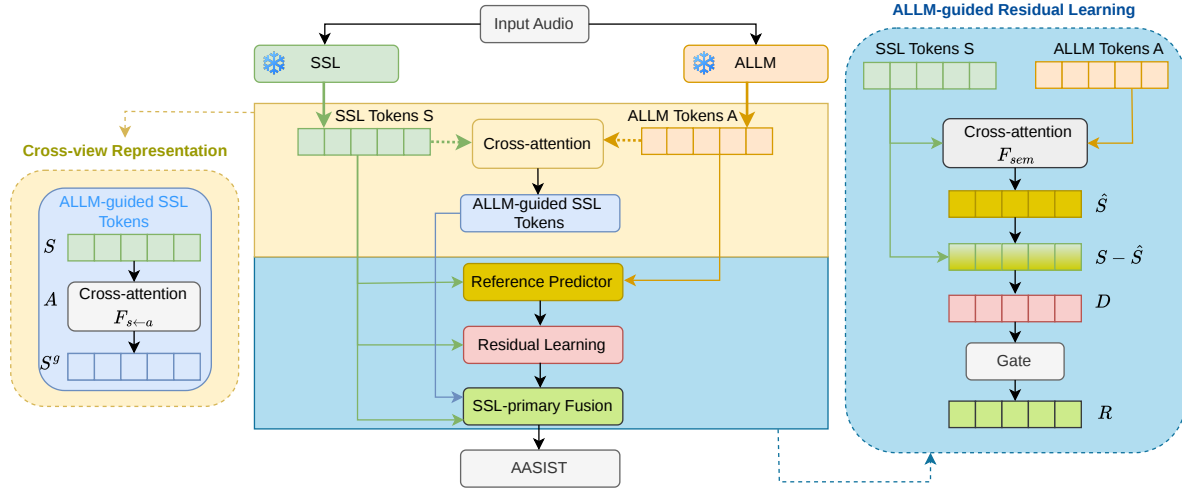


Fig. 1. Overview of CRAF with ALLM-guided representation learning, cross-view residual learning, and SSL-primary fusion.

guided cross-view residual learning to capture complementary SSL information insufficiently represented by the ALLM view. Third, we develop SSL-primary adaptive fusion to selectively integrate guided and residual information while preserving the original SSL representation.

2. PROPOSED METHOD

The overall architecture of the proposed CRAF framework is illustrated in Fig. 1. The framework begins with frozen SSL and ALLM encoders that extract complementary fine-grained acoustic and high-level contextual representations, respectively. ALLM-guided cross-view attention then enriches the SSL stream, while an SSL reference estimator derives an ALLM-conditioned reference to isolate complementary SSL information. The resulting cross-view residual is selectively refined and combined with the guided SSL representation through SSL-primary adaptive fusion before AASIST-based classification.

Cross-view Representation. Given an input waveform x , frozen SSL and ALLM encoders extract complementary representations, which are projected into a shared d -dimensional latent space:

$$\mathbf{S} = P_s(E_{\text{SSL}}(x)), \quad \mathbf{A} = P_a(E_{\text{ALLM}}(x)), \quad (1)$$

where $E_{\text{SSL}}(\cdot)$ and $E_{\text{ALLM}}(\cdot)$ denote the pretrained SSL and ALLM encoders, respectively, and $P_s(\cdot)$ and $P_a(\cdot)$ are trainable projection layers. The projected representations are $\mathbf{S} \in \mathbb{R}^{T_s \times d}$ and $\mathbf{A} \in \mathbb{R}^{T_a \times d}$, where T_s and T_a denote their sequence lengths. We set $d = 256$, while both pretrained encoders remain frozen throughout training.

To inject high-level ALLM information into the temporally fine-grained SSL stream, we apply ALLM-guided cross-view attention:

$$\mathbf{S}^g = F_{s \leftarrow a}(\mathbf{S}, \mathbf{A}), \quad (2)$$

where $F_{s \leftarrow a}(\cdot)$ uses $\mathbf{S}$ as queries and $\mathbf{A}$ as keys and values. The resulting $\mathbf{S}^g \in \mathbb{R}^{T_s \times d}$ preserves the temporal resolution

of the SSL stream while incorporating contextual information from the ALLM representation.

ALLM-guided Residual Learning. Cross-view attention enriches the SSL stream with ALLM context, but does not explicitly separate information already represented by the ALLM view from complementary SSL information. CRAF addresses this by independently estimating an ALLM-conditioned SSL reference and decomposing the SSL representation through a cross-view residual:

$$\begin{aligned} \hat{\mathbf{S}} &= F_{\text{ref}}(\mathbf{S}, \mathbf{A}), \\ \mathbf{D} &= \text{LN}(\mathbf{S} - \hat{\mathbf{S}}), \end{aligned} \quad (3)$$

where $F_{\text{ref}}(\cdot)$ is an independently parameterized stack of cross-attention layers using $\mathbf{S}$ as queries and $\mathbf{A}$ as keys and values, and $\text{LN}(\cdot)$ denotes layer normalization. Although $F_{s \leftarrow a}$ and F_{ref} share the same cross-attention formulation, they serve distinct functions: $F_{s \leftarrow a}$ transfers contextual information into the SSL stream, whereas F_{ref} estimates the portion of the SSL representation explained by the ALLM view. Consequently, $\mathbf{D} \in \mathbb{R}^{T_s \times d}$ isolates complementary SSL information relative to the ALLM-conditioned reference.

To retain only detection-relevant residual information, CRAF applies a feature-wise residual gate conditioned jointly on the original SSL representation, the estimated reference, and their residual:

$$\mathbf{R} = F_{\text{enh}} \left(\sigma \left(F_r \left([\mathbf{S}, \hat{\mathbf{S}}, \mathbf{D}] \right) \right) \odot \mathbf{D} \right), \quad (4)$$

where $[\cdot]$ denotes feature-wise concatenation, $F_r(\cdot)$ is a feed-forward gating network, $\sigma(\cdot)$ is the sigmoid function, $\odot$ denotes element-wise multiplication, and $F_{\text{enh}}(\cdot)$ is a feed-forward refinement network. The resulting $\mathbf{R} \in \mathbb{R}^{T_s \times d}$ represents the selectively refined cross-view residual.

SSL-Primary Adaptive Fusion. CRAF combines the original SSL representation $\mathbf{S}$, the ALLM-guided SSL representation $\mathbf{S}^g$, and the refined residual $\mathbf{R}$ while explicitly preserving

Table 1. Number of utterances in the ASVspoof 5 dataset.

Dataset	Splits	Number of Utterances		
		Genuine	Spoofed	Total
ASVspoof 5	Training	18,797	163,560	182,357
	Development	31,334	109,616	140,950
	Evaluation	138,688	542,086	680,774

SSL as the primary stream. Feature-wise fusion weights are computed as

$$\mathbf{G}_f = \tanh(F_f([\mathbf{S}, \mathbf{S}^g, \mathbf{R}])), \quad (5)$$

where $F_f(\cdot)$ denotes the fusion network. The final fused representation is

$$\mathbf{Z} = \mathbf{S} + \text{Dropout}(\mathbf{G}_f \odot (\mathbf{R} + \lambda \mathbf{S}^g)), \quad (6)$$

where λ controls the contribution of the ALLM-guided SSL representation. This formulation preserves the original SSL representation through a residual path while adaptively injecting contextual information from $\mathbf{S}^g$ and complementary cross-view information from $\mathbf{R}$.

Training Objective. Given the fused representation $\mathbf{Z}$, the AASIST-based classifier produces posterior probabilities $\mathbf{p}_i = [p_{i,1}, p_{i,2}]$ for bonafide and deepfake speech. The model is optimized using weighted cross-entropy with label smoothing:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{B} \sum_{i=1}^B \sum_{c=1}^2 w_c \tilde{y}_{i,c} \log p_{i,c}, \quad (7)$$

where B is the mini-batch size, w_c denotes the class-specific weight, $\tilde{y}_{i,c}$ is the label-smoothed target, and $p_{i,c}$ is the predicted posterior probability for class c . The projection, cross-view attention, reference estimation, residual learning, adaptive fusion, and classification modules are jointly optimized, while the pretrained SSL and ALLM encoders remain frozen.

3. EXPERIMENTS

3.1. Dataset and evaluation metrics

We used the ASVspoof 5 dataset [17] to evaluate the performance of the proposed method, as shown in Table 1. We use the equal error rate (EER) and the minimum detection cost function (minDCF) as evaluation metrics.

3.2. Experimental Setup

The experimental setup includes audio preprocessing, frozen pretrained encoders, CRAF modules, an AASIST-style backend, and training details. All audio is resampled to 16 kHz and normalized. We use XLS-R 300M¹ as the SSL encoder and Step-Audio², Qwen-Audio³, and Kimi-Audio⁴ as the ALLM encoder. Designed for broad audio and speech understanding, these ALLMs provide higher-level semantic and contextual representations complementary to the acoustic representations captured by SSL models. The SSL and ALLM representations are projected into a shared 256-dimensional space. The cross-view attention and reference estimator each use 2 layers, 8 heads, a feed-forward dimension of 1024, and

Table 2. Comparison with recent methods and our baselines on ASVspoof 5 Track 1 under the open condition.

Category	Method	EER (%)		minDCF
		Dev	Eval	
<i>Recent single-system methods</i>	ASTDT [18]	–	6.94	–
	ProSDD [19]	–	7.38	–
	BiCrossMamba-ST [20]	–	29.53	0.6884
	Fused SSL + NeXt-TDNN [21]	–	7.23	–
	openSMILE [22]	–	15.70	–
	C-MCSS-Mamba [23]	–	9.67	–
<i>Our baselines</i>	XLS-R + AASIST	1.22	9.78	0.1952
	Step-Audio + AASIST	7.40	11.33	0.2223
	Qwen-Audio + AASIST	6.76	12.37	0.2473
	Kimi-Audio + AASIST	3.93	7.35	0.1437
<i>Our proposed</i>	CRAF (Step-Audio)	4.65	8.45	0.1679
	CRAF (Qwen-Audio)	4.20	8.07	0.2033
	CRAF (Kimi-Audio)	2.41	5.96	0.1192

0.1 dropout. Residual gating and SSL-primary fusion operate in the same latent space. The AASIST-style backend uses 2 graph layers with hidden dimension 128, pooling ratio 0.7, and dropout 0.2. Models are trained with AdamW (2×10^{-5} learning rate, 10^{-3} weight decay, bfloat16) using weighted cross-entropy with 0.05 label smoothing, gradient accumulation of 4, and gradient clipping at 5.0. The learning rate is halved on development EER plateaus, and the best development checkpoint is used for evaluation.

3.3. Results

The results in Table 2 demonstrate the effectiveness of CRAF across different ALLM encoders. All CRAF variants improve over their corresponding single-encoder baselines on the evaluation set, indicating that combining fine-grained SSL representations with higher-level ALLM information provides complementary benefits for deepfake speech detection. Among the evaluated configurations, CRAF (Kimi-Audio) achieves the best performance, with an evaluation EER of 5.96% and a minDCF of 0.1192, improving over both the Kimi-Audio and XLS-R baselines.

Although XLS-R + AASIST obtains the lowest development EER, its performance degrades considerably on the evaluation set. In contrast, CRAF shows stronger evaluation performance, suggesting improved robustness to unseen spoofing conditions. Moreover, the consistent gains observed with Step-Audio, Qwen-Audio, and Kimi-Audio indicate that the effectiveness of CRAF is not dependent on a specific ALLM encoder, but instead results from the proposed cross-view interaction and residual-aware fusion.

CRAF (Kimi-Audio) also achieves the best evaluation EER among the recent single-system methods considered in this comparison. Notably, CRAF is evaluated as a single model without model- or score-level ensembling. Overall, these results support the effectiveness of CRAF for robust deepfake speech detection under unseen conditions.

¹XLS-R checkpoint, ²Step-Audio-2-mini-Base, ³Qwen2-Audio-7B-Instruct, ⁴Kimi-Audio-7B-Instruct.

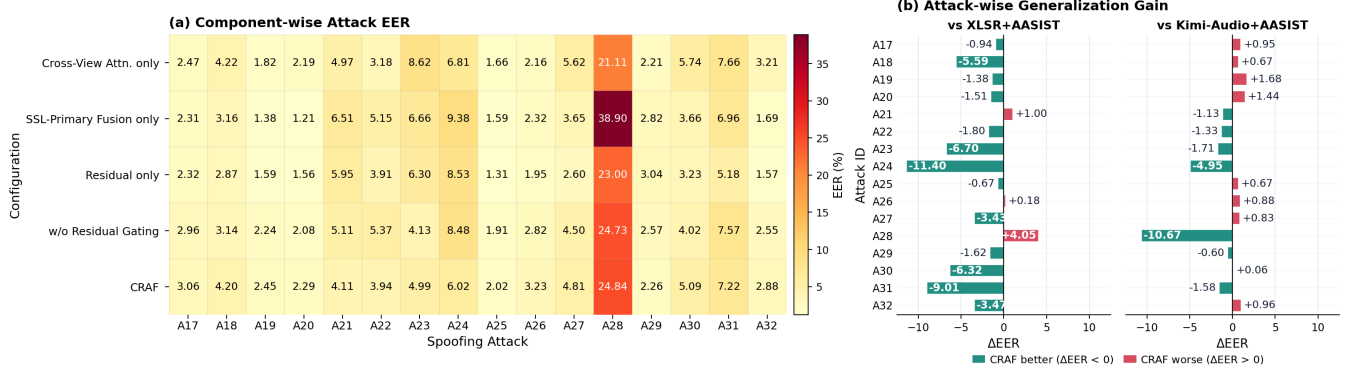


Fig. 2. Attack-wise analysis on ASVspoof 5 eval. (a) EER (%) across CRAF configurations. (b) Δ EER of CRAF relative to XLS-R+AASIST and Kimi-Audio+AASIST.

Table 3. Ablation study of CRAF components.

Category	Method	EER (%)		minDCF Eval
		Dev	Eval	
XLS-R	AASIST	1.22	9.78	0.1952
Step-Audio	AASIST	7.40	11.33	0.2223
Qwen-Audio	AASIST	6.76	12.37	0.2473
Kimi-Audio	AASIST	3.93	7.35	0.1437
Step-Audio	SSL-Primary Fusion only	6.58	8.68	0.1675
	Cross-view Attention only	2.87	7.98	0.1593
	Residual only	6.26	9.41	0.1876
Qwen-Audio	SSL-Primary Fusion only	4.29	8.35	0.1666
	Cross-view Attention only	3.04	8.41	0.1678
	Residual only	4.97	8.59	0.1708
Kimi-Audio	SSL-Primary Fusion only	2.70	7.12	0.1411
	Cross-view Attention only	1.88	6.71	0.1342
	Residual only	1.99	6.65	0.1328
	w/o Residual Gating	3.01	6.69	0.1335
Kimi-Audio	CRAF	2.41	5.96	0.1192

3.4. Ablation Studies

The ablation results in Table 3 show that cross-view attention, residual learning, and SSL-primary fusion each contribute to the effectiveness of CRAF across different ALLM encoders. With Kimi-Audio, the full CRAF achieves the best performance, reaching an EER of 5.96% and a minDCF of 0.1192, outperforming all individual component variants. The consistent degradation observed when using only one component indicates that these modules provide complementary benefits rather than redundant functionality. Removing residual gating also increases the EER to 6.69% and the minDCF to 0.1335, confirming the importance of selectively refining residual information. Overall, the consistent trends in both EER and minDCF demonstrate that jointly integrating the three components improves robustness to unseen conditions.

3.5. Attack-wise Analysis

The individual CRAF configurations exhibit distinct behaviors across spoofing attacks, as shown in Fig. 2(a). No single component is consistently optimal across all attacks: the full CRAF achieves the lowest EER among the compared configurations on A21 and A24, while individual components

perform better on several other attacks. A28 remains particularly challenging, with EERs ranging from 21.11% to 38.90%. These variations show that ALLM-guided cross-view attention, residual learning, and SSL-primary fusion respond differently to individual spoofing attacks, suggesting that each component contributes distinct attack-dependent information and plays a complementary role within the full framework.

CRAF also exhibits stronger attack-wise generalization than the single-view baselines, as illustrated in Fig. 2(b). Compared with XLS-R+AASIST, CRAF reduces EER on 13 of 16 attacks, with the largest improvements on A24 (−11.40), A31 (−9.01), A23 (−6.70), A30 (−6.32), and A18 (−5.59) percentage points. Compared with Kimi-Audio+AASIST, the gains are more attack-dependent, with improvements on 7 of 16 attacks, most notably A28 (−10.67) and A24 (−4.95). These results show that CRAF provides substantial gains over the SSL-only baseline while improving the stronger ALLM-only baseline on specific attacks, demonstrating that jointly exploiting both representation views improves robustness across diverse unseen spoofing attacks.

4. CONCLUSION

This work proposed CRAF, a cross-view residual-aware fusion framework for deepfake speech detection that jointly exploits complementary SSL and ALLM representations. By using the ALLM representation as a high-level reference, CRAF preserves complementary SSL residual information and adaptively integrates it with ALLM-guided SSL features. Experiments on ASVspoof 5 showed improved performance over the individual encoders and direct fusion baseline, demonstrating the effectiveness of CRAF for robust detection of diverse and unseen spoofing attacks. Future work will investigate improved cross-view interaction and generalization across different deepfake datasets.

5. REFERENCES

- [1] Jaehyeon Kim, Jungil Kong, and Juhee Son, “Conditional variational autoencoder with adversarial learning for end-to-end text-to-speech,” in *Proc. ICML*, 2021, pp. 5530–5540.
- [2] Massimiliano Todisco, Xin Wang, Ville Vestman, Md Sahidullah, Héctor Delgado, Andreas Nautsch, Junichi Yamagishi, Nicholas Evans, Tomi Kinnunen, and Kong Aik Lee, “Asvspoof 2019: Future horizons in spoofed and fake audio detection,” in *Interspeech 2019*, 2019.
- [3] Massimiliano Todisco, Héctor Delgado, and Nicholas Evans, “A new feature for automatic speaker verification anti-spoofing: Constant q cepstral coefficients,” in *Proc. Odyssey 2016*, 2016, pp. 283–290.
- [4] Hemlata Tak, Jose Patino, Massimiliano Todisco, Andreas Nautsch, Nicholas Evans, and Anthony Larcher, “End-to-end anti-spoofing with rawnet2,” in *Proc. ICASSP*, 2021, pp. 6369–6373.
- [5] Jee-weon Jung, Hee-Soo Heo, Hemlata Tak, Hye-jin Shim, Joon Son Chung, Bong-Jin Lee, Ha-Jin Yu, and Nicholas Evans, “Aasist: Audio anti-spoofing using integrated spectro-temporal graph attention networks,” in *Proc. ICASSP*, 2022, pp. 6367–6371.
- [6] Tianxiang Chen, Avrosh Kumar, Parav Nagarsheth, Ganesh Sivaraman, and Elie Khoury, “Generalization of audio deepfake detection,” in *Proc. Odyssey 2020*, 2020, pp. 132–137.
- [7] Junichi Yamagishi, Xin Wang, Massimiliano Todisco, Md Sahidullah, Jose Patino, Andreas Nautsch, Xuechen Liu, Kong Aik Lee, Tomi Kinnunen, Nicholas Evans, and Héctor Delgado, “Asvspoof 2021: Accelerating progress in spoofed and deepfake speech detection,” in *Proc. ASVspoof*, 2021, pp. 47–54.
- [8] Octavian Pascu, Adriana Stan, Dan Oneata, Elisabeta Oneata, and Horia Cucu, “Towards generalisable and calibrated audio deepfake detection with self-supervised representations,” in *Proc. Interspeech*, 2024, pp. 4828–4832.
- [9] Zhiyong Wang, Ruibo Fu, Zhengqi Wen, Jianhua Tao, Xiaopeng Wang, Yuankun Xie, Xin Qi, Shuchen Shi, Yi Lu, Yukun Liu, et al., “Mixture of experts fusion for fake audio detection using frozen wav2vec 2.0,” in *Proc. ICASSP*, 2025, pp. 1–5.
- [10] Lanting Li, Tianliang Lu, Xingbang Ma, Mengjiao Yuan, and Da Wan, “Voice deepfake detection using the self-supervised pre-training model hubert,” *Applied Sciences*, vol. 13, no. 14, pp. 8488, 2023.
- [11] Hao Gu, Jiangyan Yi, Chenglong Wang, Jianhua Tao, Zheng Lian, Jiayi He, Yong Ren, Yujie Chen, and Zhengqi Wen, “Allm4add: Unlocking the capabilities of audio large language models for audio deepfake detection,” in *Proc. ACMMM*, 2025, pp. 11736–11745.
- [12] Akanksha Chuchra, Shukesh Reddy, Sudeepta Mishra, Abhijit Das, and Abhinav Dhall, “Investigating the viability of employing multi-modal large language models in the context of audio deepfake detection,” in *Proc. IJCB*, 2025, pp. 1–11.
- [13] Xiaoxuan Guo, Yuankun Xie, Haonan Cheng, Jiayi Zhou, Jian Liu, Hengyan Huang, Long Ye, and Qin Zhang, “Towards explicit acoustic evidence perception in audio llms for speech deepfake detection,” *arXiv preprint arXiv:2601.23066*, 2026.
- [14] Yassine El Kheir, Xin Wang, Wanqing Ge, Tim Polzehl, Sebastian Moeller, and Junichi Yamagishi, “Textual acoustic grounding for generalizable llm-based deepfake voice detection,” *arXiv preprint arXiv:2608.30622*, 2026.
- [15] Dengtai Tan, Boao Tan, Deyi Yang, Yang Yang, and Chengyu Niu, “Mutual-guidance framework for audio deepfake detection via multi-dimensional feature interaction,” *Neurocomputing*, vol. 672, pp. 132732, 2026.
- [16] Dan Li, Haibao Wang, Yufeng Wang, and Shengpei Wang, “Instance-wise multi-view representation learning,” *Information Fusion*, vol. 91, pp. 612–622, 2023.
- [17] Xin Wang, Héctor Delgado, Hemlata Tak, Jee-weon Jung, Hye-jin Shim, Massimiliano Todisco, Ivan Kukanov, Xuechen Liu, Md Sahidullah, Tomi H Kinnunen, et al., “Asvspoof 5: crowdsourced speech data, deepfakes, and adversarial attacks at scale,” in *Proc. ASVspoof 2024*, 2024, pp. 1–8.
- [18] Taiba Maijd Wani, Syed Asif Ahmad Qadri, Arselan Ashraf, and Irene Amerini, “Astdt: an interpretable adaptive spectro-temporal diffusion transformer for audio deepfake detection,” *EURASIP Journal on Information Security*, vol. 2025, no. 1, pp. 32, 2025.
- [19] A. Mahapatra, I. R. Ulgen, K. A. Lee, N. Andrews, and B. Sisman, “Prosdd: Learning prosodic representations for speech deepfake detection against expressive and emotional attacks,” in *Proc. Interspeech*, 2026.
- [20] Yassine El Kheir, Tim Polzehl, and Sebastian Möller, “BiCrossMamba-ST: Speech Deepfake Detection with Bidirectional Mamba Spectro-Temporal Cross-Attention,” in *Proc. Interspeech*, 2025, pp. 2235–2239.
- [21] Gul Tahaoglu, “Robust deepfake audio detection via an improved next-tdnn with multi-fused self-supervised learning features,” *Applied Sciences*, vol. 15, no. 17, pp. 9685, 2025.
- [22] Octavian Pascu, Dan Oneată, Horia Cucu, and Nicolas Müller, “Easy, interpretable, effective: opensmile for voice deepfake detection,” in *Proc. ICASSP*, 2025, pp. 1–5.
- [23] Gaopeng Zhang, Shidong Liu, Dengtao Zhang, and Liang Tang, “C-mcss-mamba: Counterfactual mechanism-contrastive selective scan within mamba for block-causal speech deepfake detection,” *Applied Sciences*, vol. 16, no. 17, 2026.